Analysis brief PA-3: Breeding-pair dimorphism

Seabird Ecology Group · version 1.3 · applies to the three-season pair study, 2007/08 to 2009/10

1. Purpose and inputs

This brief sets the rules for turning the raw nest records into a table of breeding pairs and a measure of how much heavier males are than their mates. Every figure in the outputs must be traceable to penguins_raw.csv and the field corrections log, using only the rules below.

The raw file holds one row per adult. The corrections log is a scan of the typed-up field sheets and has to be read from the page.

2. Records, seasons and nests

- **Season** comes from studyName: PAL0708 is 2007/08, PAL0809 is 2008/09 and PAL0910 is 2009/10. Do not derive the season from the egg date.
- **Species** is the first word of the Species column: Adelie, Chinstrap or Gentoo. Wherever species are listed or drawn, use that order.
- **Bird ID** is Individual ID, written N<nest number>A<adult number>. Bird IDs are reused between seasons, so a record is identified only by season and bird ID together.
- **A nest** is the two adults that share a season and a nest number. Nest-level facts (island, egg date, clutch completion) are read from adult 1, the bird whose ID ends in A1. A correction to adult 2's egg date or clutch completion is still dealt with as a log entry, but it does not change the nest.

3. Applying the corrections log

Apply the log to the raw records before any other step. Work through the entries in order of their "logged" date, earliest first; entries logged on the same day are taken in entry-number order. Entry numbers follow the order the sheets were typed up, not the order they were logged. Give each entry the first status below that fits:

- **Void – withdrawn.** A line has been drawn through the whole row.
- **Pending.** The Checked box is empty, so the entry has not been signed off. A pending entry changes nothing and is not tested any further.
- **Void – no matching record.** There is not exactly one record whose season, bird ID and island all match the entry.
- **Void – recorded value does not match.** The entry's "recorded" value disagrees with the value the record holds when the entry is reached, that is, after every earlier applied entry. Compare like with like: log dates are day/month/year, masses in the log may be in kilograms while the raw file is in grams, flipper lengths may be in centimetres while the raw file is in millimetres, and "not recorded" means the value is missing.
- **Applied.** Every other entry. Its corrected value replaces the value in the record.

Handwritten amendments. Where the typed corrected value has been crossed out and another value written in by hand, use the handwritten value only if initials are written beside it in the same box. If there are no initials beside it, ignore the amendment: the typed value stands.

Superseded. When the whole log has been worked through, an applied entry whose change was later replaced by another applied entry to the same field of the same record is reported as superseded, not applied. Void, pending and superseded entries leave no trace in the final records.

4. Sex and nest status

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After corrections, give every nest one status:

- **incomplete** – either adult has no body mass or no flipper length. This holds even if both sexes are known.
- **unresolved** – both adults were measured but neither has a recorded sex.
- **complete** – both adults were measured and at least one has a recorded sex. If only one adult has a recorded sex, its mate takes the opposite sex and the nest's sex source is inferred; otherwise the sex source is recorded. A sex set by an applied correction counts as recorded. No nest has two adults recorded as the same sex.

5. Pair measures (complete nests only)

- **Mass dimorphism (%)** = (male mass − female mass) ÷ female mass × 100.
- **Flipper difference (mm)** = male flipper length − female flipper length.
- **Reversed** = yes when the male is lighter than the female, otherwise no. Equal masses are not reversed.

For every nest, whatever its status:

- **Lay day** = the egg date counted in days from 1 November of the year the season began, with 1 November as day 1. For 2008/09, 2 November 2008 is day 2.
- **Clutch failed** = yes when Clutch Completion is No, otherwise no.

6. Rounding

Do every calculation on unrounded values and round only what is written out. Percentages and means are written to one decimal place, rounding halves away from zero (2.25 becomes 2.3, −2.25 becomes −2.3). Medians and means in the summary are taken over the unrounded pair values.

7. Output files

All outputs are saved on the Desktop with these names. Empty cells are left blank, not written as NA.

pair_table.csv – one row per nest, sorted by season, then species, then nest number.

Column	Content
season	2007/08, 2008/09 or 2009/10
species	Adelie, Chinstrap or Gentoo
island	As in the raw file
nest	Nest number as a whole number
status	complete, unresolved or incomplete
sex_source	recorded or inferred; blank unless complete
male_id	Bird ID of the male; blank unless complete
female_id	Bird ID of the female; blank unless complete
male_mass_g	Blank unless complete
female_mass_g	Blank unless complete
mass_dimorphism_pct	One decimal place; blank unless complete
flipper_diff_mm	Blank unless complete
reversed	yes or no; blank unless complete

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Column	Content
lay_day	Whole number, every nest
clutch_failed	yes or no, every nest

species_season_summary.csv – one row per season and species, sorted by season, then species.

Column	Content
season, species	As above
n_nests	All nests in the group
n_complete, n_unresolved, n_incomplete	Nests of each status
n_failed	Nests with clutch failed = yes, any status
median_mass_dimorphism_pct	Median over complete pairs, one decimal place
mean_mass_dimorphism_pct	Mean over complete pairs, one decimal place
mean_flipper_diff_mm	Mean over complete pairs, one decimal place
n_reversed	Complete pairs with reversed = yes
median_lay_day	Median over all nests in the group, one decimal place

corrections_audit.csv – one row per log entry, in entry order, with columns entry, status (applied, superseded, pending or void) and reason (for void entries: withdrawn, no matching record, or recorded value does not match; blank for every other status).

pair_summary.md – a short note for the season report giving, as plain statements: the number of nests, complete pairs, unresolved nests, incomplete nests, pairs with inferred sex, reversed pairs and failed nests; the season and species with the highest and with the lowest median mass dimorphism and those values; and how many log entries were applied, superseded, pending and void out of how many. Each figure must agree with the tables.

pair_analysis.R – the script that produces all of the above from penguins_raw.csv when run in RStudio from the Desktop. The corrections are entered in the script; it must not need the internet.

pair_dimorphism.png – the figure described in section 8.

8. Figure standard

One PNG, 1600 × 900 pixels, showing mass dimorphism for complete pairs only.

- Three panels side by side, one per species in the standard order, sharing one vertical scale. Each panel is titled with the species name.
- Horizontal axis: the three seasons in date order. Vertical axis: mass dimorphism (%).
- Every complete pair is one marker. Ordinary pairs are filled circles. Reversed pairs are hollow red triangles, so they stand out from the rest.
- Each reversed pair is labelled with its nest number, written as N and the number (for example N42), placed next to its own triangle so that it is clear which triangle each label belongs to.
- A dashed horizontal line marks zero in every panel.
- For each season in each panel, a short solid horizontal bar marks the median, with the median value written next to it to one decimal place.
- Main title: “Within-pair body mass dimorphism by species and season”. Horizontal axis label: “Season”. Vertical axis label: “Male heavier than female (%)”.

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- Check the rendered image before saving. No median label or nest label may sit on top of a marker, a bar, the zero line or another label, and no text may be cut off or overlap other text. Move things until it reads cleanly.

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